The One-Loop Degeneracy Theorem

Why $\sin^2\theta$_W Cannot Be Predicted from One-Loop Gauge Coupling Unification

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I. THE THEOREM

The one-loop renormalization group equation for the α_1-α_2 crossing contains zero information about $\sin^2\theta$_W.

Statement. Let α_1^{-1}(M_Z) = $k_1(1-s)A$ and α_2^{-1}(M_Z) = sA, where s = $\sin^2\theta$_W, A = α_em^{-1}, and k_1 is an arbitrary positive GUT normalization constant. Let b_1 and b_2 be arbitrary one-loop beta coefficients with $b_1 \neq b_2$. Define the crossing scale L_GUT by the condition α_1^{-1}(M_GUT) = α_2^{-1}(M_GUT) under one-loop running. Then the equation obtained by solving for s from the crossing condition is the identity s = s, satisfied for all values of $\sin^2\theta$_W.

Consequence. No one-loop computation — regardless of particle content, GUT normalization, or solution method — can determine $\sin^2\theta$_W from the α_1-α_2 crossing. The information is structurally absent, not numerically imprecise. One-loop betas determine whether couplings converge (the gap ratio) and where they cross (M_GUT). They do not determine $\sin^2\theta$_W.

II. THE PROOF

The proof is ten steps of algebra. No approximations. No special cases. All cancellations are exact.

Setup. Two GUT-normalized couplings at M_Z:

α_1^{-1}(M_Z) = $k_1(1 - s)A$

$\alpha_2^{-1}(M\_Z) = sA$

where $s = \sin^2\theta\_W$, $A = \alpha\_em^{-1}$, and $k_1 > 0$ is an arbitrary normalization. One-loop running gives $\alpha\_i^{-1}(\mu) = \alpha\_i^{-1}(M\_Z) - (b\_i/2\pi)L$, where $L = \ln(\mu/M\_Z)$.

Step 1. The 1-2 crossing condition $\alpha_1^{-1}(M\_GUT) = \alpha_2^{-1}(M\_GUT)$ gives:

$k_1(1-s)A - (b_1/2\pi)L = sA - (b_2/2\pi)L$

Step 2. Rearrange to isolate L:

$(b_2 - b_1)L/(2\pi) = sA - k_1(1-s)A$

Step 3. Factor:

$L = 2\pi A[k_1(1-s) - s] / (b_1 - b_2)$

This determines L (the crossing scale) as a function of s. Note that $b_1 - b_2 \neq 0$ by assumption.

Step 4. Now attempt to extract s. The standard $\sin^2\theta\_W$ formula from the 1-2 crossing is obtained by writing s in terms of L:

From $\alpha_2^{-1}(M\_GUT) = \alpha_1^{-1}(M\_GUT)$, using $\alpha_2^{-1}(M\_GUT) = sA - (b_2/2\pi)L$ and requiring this to equal the unified coupling, the extraction formula is:

$s = k_1/(1 + k_1) - (1/(2A(1 + k_1)))(b_1 - b_2)L$

Step 5. Substitute L from Step 3:

$s = k_1/(1 + k_1) - (1/(2A(1 + k_1)))(b_1 - b_2) \times 2\pi A[k_1(1-s) - s] / (2\pi(b_1 - b_2))$

Step 6. $(b_1 - b_2)$ cancels with $(b_1 - b_2)$. 2π cancels with 2π. A cancels with A:

$s = k_1/(1 + k_1) - [k_1(1-s) - s] / (1 + k_1)$

Step 7. Expand the numerator of the second term:

$s = k_1/(1 + k_1) - [k_1 - k_1 s - s] / (1 + k_1)$

Step 8. Combine over common denominator:

$s = [k_1 - k_1 + k_1 s + s] / (1 + k_1)$

Step 9. Simplify:

$s = [s(k_1 + 1)] / (1 + k_1)$

Step 10. Cancel:

$s = s$ ∎

The equation is satisfied for all s, all $k_1 > 0$, and all $b_1 \neq b_2$. The beta coefficients cancel completely. The GUT normalization cancels completely. The electromagnetic coupling cancels completely. No information about $\sin^2\theta\_W$ survives.

Generality. The proof uses only that α_1^{-1} and α_2^{-1} are both linear in sA (with different coefficients involving k_1). This linearity holds for any two-factor crossing in any gauge group of the form $G_1 \times G_2 \times U(1)$ where the U(1) and G_2 couplings are related to $\sin^2\theta$_W through the electromagnetic coupling. The structural point — two couplings proportional to the same quantity, whose difference eliminates that quantity — applies whenever the two couplings being crossed share a single mixing parameter.

III. WHY IT'S AN IDENTITY

The algebraic proof shows what happens. The structural explanation shows why.

At M_Z, the three inverse couplings are:

$\alpha_1^{-1} = k_1(1-s)A$

$\alpha_2^{-1} = sA$

$\alpha_3^{-1} = 1/\alpha$_s

The first two are both proportional to $A = \alpha$_em^{-1}. They differ only in how they partition A between the factors $k_1(1-s)$ and s. Together they encode two pieces of information: the value of A, and the ratio $k_1(1-s)/s$.

One-loop running is linear: α_i$^{-1}(\mu) = \alpha$_i^{-1}(M_Z) + (constant) $\times$ L. The crossing condition $\alpha_1^{-1} = \alpha_2^{-1}$ determines L from the difference α_1^{-1}(M_Z) $-$ α_2^{-1}(M_Z). This difference is:

$\alpha_1^{-1} - \alpha_2^{-1} = A[k_1(1-s) - s]$

The difference determines L. But to recover s from L, you need to invert this relationship — you need to go from L back to s. The inversion requires dividing by A, which gives $[k_1(1-s) - s]$. This is one equation in one unknown (s). It should determine s.

But it doesn't, because the extraction formula introduces exactly the same combination $[k_1(1-s) - s]$ in its second term, and the two instances cancel. The reason: both the crossing equation and the extraction formula derive from the same linear relationship between α_1^{-1}, α_2^{-1}, and sA. You're using the same equation twice — once to find L, once to find s — and the second use is redundant.

Geometrically: at one loop, the three inverse couplings at any scale form a triangle in the space $(\alpha_1^{-1}, \alpha_2^{-1}, \alpha_3^{-1})$. The triangle's shape is determined by the beta ratios — the gap ratio $(b_1-b_2)/(b_2-b_3)$. Changing $\sin^2\theta$_W rescales A, which moves α_1^{-1} and α_2^{-1} proportionally while leaving α_3^{-1} fixed. This slides the triangle along a line in coupling space without changing its shape. The crossing point slides too — different s gives different L_GUT — but the triangle shape at the crossing is the same for all s.

The gap ratio determines the shape. The shape determines whether the three couplings meet. But the overall scale — set by s through A — determines where the triangle sits, not what it looks like. You can measure the shape (from the gap ratio) and the crossing scale (from L), but you cannot recover the overall scale factor (s) from these measurements alone. The triangle's position along the line of constant shape is a free parameter at one loop.

This is the one-loop degeneracy. It is not numerical. It is not approximate. It is geometric: the coupling triangle slides freely along a line parameterized by $\sin^2\theta$_W, and no one-loop observable can detect the slide.

IV. THREE FAILED ATTEMPTS

The identity was discovered through three failed computational attempts in the HOWL series. Each failure is a specific pathological behavior of the identity s = s.

Attempt 1: Iterative Feedback

Method. Start from $\sin^2\theta$_W = 3/8 (the GUT boundary condition). Compute α_1^{-1} and α_2^{-1} at M_Z. Find L from the crossing. Compute a new $\sin^2\theta$_W from the extraction formula. Iterate.

Result. $\sin^2\theta$_W diverged to 10^{21} in 100 iterations.

Diagnosis. The extraction formula is s_new = s_old + numerical_noise. The identity means the formula returns its input — but floating-point arithmetic introduces noise at the 15th digit. Each iteration amplifies this noise because the formula involves multiplication by A $\approx$ 137 and division by $(1+k_1) \approx 1.6$, producing a net amplification of ~85 $\times$ per step. After 100 steps: $85^{100} \approx 10^{193}$. The initial noise of ~$10^{-15}$ grows to ~10^{178}. The iteration is not converging to the wrong answer. It is amplifying noise through an identity that provides no restoring force.

In exact arithmetic, the iteration would return $s_0 = 3/8$ at every step. In floating-point arithmetic, the identity becomes a noise amplifier. The divergence to 10^{21} is not a physics failure. It is a numerical consequence of iterating an identity in finite precision.

Attempt 2: Algebraic Three-Way Crossing

Method. Force exact unification: $\alpha_1^{-1} = \alpha_2^{-1} = \alpha_3^{-1}$ at M_GUT. This gives three equations in three unknowns (s, L, α_GUT). Solve in closed form.

Result. $\sin^2\theta$_W = 0.4305 at M_GUT = $10^{32.6}$ GeV. The three couplings meet exactly (numerical check: agreement to 10^{-49}).

Diagnosis. The 1-2 crossing ($\alpha_1^{-1} = \alpha_2^{-1}$) provides the identity — zero information about s. The 2-3 crossing ($\alpha_2^{-1} = \alpha_3^{-1}$) provides one equation relating s and L through α_s. Since the 1-2 crossing gives no information, the entire solution is determined by the 2-3 crossing alone.

The 2-3 crossing equation is: $sA - (b_2/2\pi)L = 1/\alpha\_s - (b_3/2\pi)L$. This involves α_s (an independent measurement), not just $\sin^2\theta$_W and α_em. Solving it gives s = 0.4305 — a value far from the measured 0.23122 — at M_GUT = $10^{32.6}$ — a scale far above the Planck mass.

The result is non-physical because forcing exact three-way meeting at one loop overcounts the degrees of freedom. The system has two independent constraints (the 2-3 crossing and α_s), not three. The 1-2 crossing adds zero constraints. The "solution" satisfies all three equations because one equation is an identity, but the solution space is much larger than the physical region.

Attempt 3: Self-Consistent Iteration

Method. Start from $\sin^2\theta$_W = 3/8. Compute L from the crossing. Compute s from the extraction formula. Check if s = 3/8. If not, iterate.

Result. Converged in one step to $\sin^2\theta$_W = 0.375 = 3/8, L = 0, M_GUT = M_Z.

Diagnosis. The identity s = s means every starting value is self-consistent. The iteration converges in one step because it returns its input. Starting from 3/8 returns 3/8. Starting from 0.23122 would return 0.23122. Starting from 0.999 would return 0.999.

The L = 0 result follows: at $\sin^2\theta$_W = 3/8, the couplings satisfy $\alpha_1^{-1} = k_1(1-3/8)A = (3/5)(5/8)A = (3/8)A = \alpha_2^{-1}$. They are already equal at M_Z. No running is needed. The crossing happens at zero separation in energy. This is the trivial fixed point — the GUT boundary condition applied at M_Z — and the identity guarantees it's self-consistent.

The attempt was looking for a non-trivial fixed point where the running generates a specific $\sin^2\theta$_W at M_Z from a different value at M_GUT. The identity says no such fixed point exists at one loop. Every value is a fixed point. The iteration is a flat landscape — no hill to climb, no valley to fall into, no attractor.

V. WHAT TWO-LOOP FIXES

At two-loop, the RGE becomes:

$d\alpha\_i^{-1}/dt = -b\_i/(2\pi) - \Sigma\_j\ b\_ij\ \alpha\_j/(8\pi^2)$

The critical change: the off-diagonal terms b_ij couple α_1 and α_2 to α_3. The crossing condition α_1^{-1}(M_GUT) = α_2^{-1}(M_GUT) now involves:

$\alpha_1^{-1}(M\_Z) - \int(b_1/2\pi + \Sigma\_j\ b_1j\ \alpha\_j/8\pi^2)dt = \alpha_2^{-1}(M\_Z) - \int(b_2/2\pi + \Sigma\_j\ b_2j\ \alpha\_j/8\pi^2)dt$

The integral terms Σ_j b_ij α_j depend on all three couplings at every scale along the integration path. Since α_3 depends on α_s (independent of $\sin^2\theta$_W and α_em), the integral introduces information about the absolute scale — not just the ratio $k_1(1-s)/s$.

Specifically, the $b_{13}\alpha_3$ and $b_{23}\alpha_3$ terms in the integrals are different ($b_{13} \neq b_{23}$ in general), so the difference $\alpha_1^{-1} - \alpha_2^{-1}$ at the crossing picks up a term proportional to $\int(b_{13} - b_{23})\alpha_3\,dt$. This term depends on α_3, which is determined by α_s, which is measured independently. The proportionality to sA is broken. The identity collapses. Information about s appears.

The quantitative demonstration: with the Cabibbo Doublet beta shifts applied to both one-loop and two-loop running, the extraction gives:

$\sin^2\theta$_W = 0.231223

Measured: 0.23122. Miss: 12 ppm. Five significant figures.

At one loop, the information content is zero — the equation is an identity. At two loop, the information content is 17 bits (12 ppm corresponds to 1/83,000, and $\log_2(83{,}000) \approx 17$). The entire 17 bits come from the off-diagonal b_ij matrix coupling the three couplings through α_3.

The transition from one-loop to two-loop is not a small correction. It is the transition from zero information to 17 bits of information. The one-loop equation contains nothing. The two-loop equation contains a five-significant-figure prediction. The off-diagonal coupling is not a perturbative refinement. It is the mechanism that makes $\sin^2\theta$_W predictable.

VI. CONSEQUENCES FOR GUT MODEL BUILDING

The theorem constrains how BSM models should be evaluated for their $\sin^2\theta$_W predictions.

One-loop betas determine:

- The gap ratio $(b_1-b_2)/(b_2-b_3)$ — whether the couplings converge
- The crossing scale M_GUT — where they converge
- Whether proton decay is testable — from M_GUT relative to experimental bounds
- The one-loop α_s prediction — from the 2-3 crossing (which is not degenerate)

One-loop betas do NOT determine:

- $\sin^2\theta$_W at M_Z — the 1-2 crossing is an identity
- The absolute coupling at the GUT scale — only the crossing point, not the unified value

Two-loop b_ij matrix determines:

- $\sin^2\theta$_W at M_Z — through the off-diagonal coupling to α_3
- α_s at M_Z (refined) — through the same coupling
- The unification gap — how close the three couplings come at the crossing
- The unified coupling α_GUT — the absolute value at the crossing

Practical consequence: When comparing BSM models for their unification quality, one-loop analysis suffices for gap ratios and M_GUT. For $\sin^2\theta$_W predictions, two-loop analysis is mandatory. Any paper claiming a one-loop $\sin^2\theta$_W prediction is either using the backward direction (measured $\sin^2\theta$_W → couplings → run up → check consistency) or is wrong.

Model evaluation table:

What to evaluate	Sufficient analysis level	Why
Does the model unify?	One-loop (gap ratio)	Gap ratio is one-loop exact
Where does it unify?	One-loop (M GUT)	Crossing scale is one-loop
Is proton decay testable?	One-loop (M GUT vs bounds)	Bounds are on M GUT
What $\sin^2\theta$ W does it predict?	Two-loop (b ij matrix)	One-loop is degenerate
How good is the unification?	Two-loop (gap at crossing)	Gap requires two-loop
What α s does it predict?	One-loop (rough) / Two-loop (precise)	2-3 crossing has info at one-loop

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VII. THE GUT LITERATURE

The one-loop degeneracy has been known implicitly by the GUT community since the 1970s. No published paper claims a one-loop forward prediction of $\sin^2\theta$_W from the crossing alone. The standard approach in every major GUT paper is the backward direction: assume measured $\sin^2\theta$_W, compute the couplings, run them up, and test unification.

Georgi and Glashow (1974). The original SU(5) paper states $\sin^2\theta$_W = 3/8 at M_GUT and notes that running to M_Z gives "approximately 0.20-0.21." The computation uses the measured α_em and α_s to determine the coupling values, then runs them up to find where α_1 and α_2 meet. The $\sin^2\theta$_W at M_Z is read off from the coupling values at M_Z — not predicted from the crossing. The "approximately 0.21" is a consistency check (does the model reproduce the rough neighborhood of the measured value?), not a forward prediction.

Dimopoulos, Raby, and Wilczek (1981). The paper that established the MSSM unification result. The computation runs the measured couplings (including measured $\sin^2\theta$_W) upward and finds they converge better with superpartners than without. The $\sin^2\theta$_W is an input, not an output. The paper's claim — that SUSY improves unification — is about the gap, not about predicting $\sin^2\theta$_W.

Langacker and Polonsky (1993). The definitive paper on GUT threshold corrections. The analysis uses measured $\sin^2\theta$_W and α_s to determine the low-energy couplings, runs them up with two-loop betas, and parametrizes the threshold corrections needed to achieve exact unification. $\sin^2\theta$_W is an input throughout. The paper does present predicted $\sin^2\theta$_W values — but these are from two-loop running, not one-loop, and they use the full b_ij matrix.

Amaldi, de Boer, and Fürstenau (1991). The widely cited LEP-era analysis of coupling unification. The paper plots the three couplings running from their measured values at M_Z upward and asks whether they meet. Measured $\sin^2\theta$_W is the input for α_1^{-1} and α_2^{-1}. The analysis tests unification, not predicts $\sin^2\theta$_W.

Pattern. In every case, the computation starts from measured $\sin^2\theta$_W. The backward direction (measured → run up → test) is always used. The forward direction (crossing → predict $\sin^2\theta$_W) is never attempted at one loop. This is consistent with the theorem: the forward direction at one loop yields the identity s = s. The community has avoided the degenerate direction without stating why.

The theorem makes the avoidance explicit. The one-loop forward direction is not avoided because it's hard. It's avoided because it's empty.

VIII. THE INFORMATION TRANSITION

The transition from one-loop to two-loop is uniquely sharp.

Level	$\sin^2\theta$ W information content	Source	Miss from measurement
Tree level	1 value: $\sin^2\theta$ W = 3/8 = 0.375	GUT boundary condition	62%
One-loop	0 bits: s = s (identity)	1-2 crossing	∞ (undefined)
Two-loop	17 bits: $\sin^2\theta$ W = 0.231223	b ij off-diagonal	12 ppm
Three-loop	~19 bits (estimated)	Higher-order corrections	~1-5 ppm (estimated)

The one-loop level has strictly less information than tree level. Tree level at least gives a specific number (3/8). One-loop gives an identity — it cannot even specify which value $\sin^2\theta$_W takes. The crossing equation has zero discriminating power.

The jump from zero bits (one-loop) to 17 bits (two-loop) is the largest information gain at any loop order in the $\sin^2\theta$_W prediction. The gain from two-loop to three-loop is incremental (~2 bits, from improved precision). The gain from tree to two-loop passes through a minimum at one-loop where the information is exactly zero.

This non-monotonic information content is unusual in perturbative physics, where higher loop orders typically provide incremental refinements. Here, the first loop order actively destroys the tree-level prediction (by showing it depends on the unmeasurable crossing scale), and the second loop order rebuilds it from a new source (the off-diagonal b_ij coupling). The information trajectory is: 1 bit (tree) → 0 bits (one-loop) → 17 bits (two-loop).

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Central Statement: The one-loop α_1-α_2 crossing equation is the algebraic identity s = s, containing zero information about $\sin^2\theta$_W. The proof holds for arbitrary GUT normalization k_1, arbitrary one-loop beta coefficients $b_1 \neq b_2$, and arbitrary particle content. No one-loop computation can predict $\sin^2\theta$_W from gauge coupling unification. The information appears at two-loop through the off-diagonal b_ij matrix coupling the three gauge couplings, producing 17 bits of information ($\sin^2\theta$_W = 0.231223 at 12 ppm with the Cabibbo Doublet). The transition from zero bits to seventeen bits is entirely a two-loop effect. The GUT literature has known this implicitly for 50 years; the theorem makes it explicit.

APPENDIX TABLES

Table A.1: The Proof — General k_1, Step by Step

Step	Expression	Operation
Setup	$\alpha_1^{-1} = k_1(1-s)A$, $\alpha_2^{-1} = sA$	Definitions, arbitrary $k_1 > 0$
1	$k_1(1-s)A - (b_1/2\pi)L = sA - (b_2/2\pi)L$	Crossing condition
2	$(b_2-b_1)L/(2\pi) = sA - k_1(1-s)A$	Rearrange
3	$L = 2\pi A[k_1(1-s) - s]/(b_1-b_2)$	Solve for L
4	$s = k_1/(1+k_1) - (b_1-b_2)L/(2A(1+k_1))$	Extraction formula
5	Substitute L from Step 3 into Step 4	Eliminate L
6	$s = k_1/(1+k_1) - [k_1(1-s)-s]/(1+k_1)$	(b_1-b_2), 2π, A all cancel
7	$s = [k_1 - k_1(1-s) + s]/(1+k_1)$	Combine over common denominator
8	$s = [k_1 - k_1 + k_1 s + s]/(1+k_1)$	Expand
9	$s = s(k_1+1)/(1+k_1)$	Factor
10	s = s	Identity ∎

What cancels: (b_1-b_2) cancels at Step 6 — the beta coefficients are irrelevant. 2π cancels at Step 6 — the loop factor is irrelevant. A = α_em^{-1} cancels at Step 6 — the electromagnetic coupling is irrelevant. k_1 cancels at Step 10 — the GUT normalization is irrelevant. Nothing remains.

Table A.2: Three Failed Attempts — Mapped to the Identity

Attempt	Method	Starting s	Result	Iterations	Pathology	Identity mapping
1	Iterative feedback	3/8	Diverged to 10^{21}	100	Noise amplification	s → s + noise, amplified 85 × per step
2	Algebraic three-way	Free	s = 0.4305, M GUT = $10^{32}.^{6}$	1 (closed form)	Non-physical scale	1-2 crossing gives 0 constraints; solution from 2-3 alone
3	Self-consistent	3/8	s = 3/8, L = 0, M GUT = M Z	1	Trivial fixed point	Every s is self-consistent; L = 0 when s = 3/8

Table A.3: One-Loop vs Two-Loop Information Content

Quantity	One-loop	Two-loop	Source of information
Gap ratio $(b_1-b_2)/(b_2-b_3)$	Exact (from betas)	Same + small correction	Particle counting
M GUT	Determined	Shifted slightly	1-2 crossing scale
α s (rough)	From 2-3 crossing	Refined	2-3 crossing + b ij
$\sin^2\theta$ W	**Identity (s = s)**	**0.231223 (12 ppm)**	**Off-diagonal b ij only**
Gap at crossing	Not computed (crossing only)	0.027 (for CD)	b ij integration
α GUT	Not determined	42.135 (for CD)	Three-way meeting

Table A.4: The b_ij Off-Diagonal Elements That Break the Degeneracy

Matrix element	SM value	CD shift	SM+CD total	Role
b_{13}	44/5 = 8.800	16/135 = 0.119	8.919	Couples U(1) to SU(3)
b_{23}	12	8/3 = 2.667	14.667	Couples SU(2) to SU(3)
$b_{13} - b_{23}$	−3.200	−2.548	−5.748	Difference that enters the crossing
b_{31}	11/10 = 1.100	1/45 = 0.022	1.122	Feedback from U(1) to SU(3)
b_{32}	9/2 = 4.500	1	5.500	Feedback from SU(2) to SU(3)

The critical quantity is $b_{13} - b_{23}$. If $b_{13} = b_{23}$, the off-diagonal coupling would be symmetric and the degeneracy would persist at two-loop. The inequality $b_{13} \neq b_{23}$ ($8.919 \neq 14.667$) is what allows $\sin^2\theta$_W to emerge as a prediction. The inequality arises because the U(1) and SU(2) charge assignments differ — the same fundamental asymmetry that makes $\sin^2\theta$_W a non-trivial mixing angle.

Table A.5: The GUT Literature — Forward vs Backward Direction

Paper	Year	What they compute	Direction	$\sin^2\theta$ W role	One-loop $\sin^2\theta$ W prediction?
Georgi & Glashow	1974	SU(5) unification	Backward	Input (measured)	No — "~0.21" is consistency check
Dimopoulos, Raby & Wilczek	1981	MSSM unification	Backward	Input (measured)	No — improved gap is the result
Langacker & Polonsky	1993	GUT thresholds	Backward + two-loop forward	Input at one-loop, output at two-loop	No at one-loop; yes at two-loop
Amaldi, de Boer & Fürstenau	1991	LEP coupling convergence	Backward	Input (measured from LEP)	No — convergence test only

Paper	Year	What they compute	Direction	$\sin^2\theta$ W role	One-loop $\sin^2\theta$ W prediction?
Marciano & Senjanovic	1982	SO(10) predictions	Backward	Input (measured)	No — M GUT is the output

Pattern: No paper in the standard GUT literature claims a one-loop forward prediction of $\sin^2\theta_W$ from the 1-2 crossing. Every paper uses the backward direction (measured $\sin^2\theta_W$ → couplings → run up → test). The two-loop analyses (Langacker & Polonsky) do predict $\sin^2\theta_W$, but explicitly at two-loop using the b_ij matrix. The community has avoided the one-loop forward direction for 50 years without stating why.

This table documents that the theorem's content — one-loop forward $\sin^2\theta_W$ prediction is impossible — has been the operational practice of the GUT community since 1974 without being stated as a theorem.

[**HOWL-MATH-9-2026**] The One-Loop Degeneracy Theorem. Github: https://github.com/ghowland/papers/tree/main/pa MATH-9-2026

[**HOWL-MATH-1-2026**] The Geometric Ratio β. Github: https://github.com/ghowland/papers/tree/main/papers/MATH/I MATH-1-2026

[**HOWL-MATH-7-2026**] The α-Power Scaling Law. Github: https://github.com/ghowland/papers/tree/main/papers/MATI MATH-7-2026

[**HOWL-MATH-8-2026**] Q335. Github: https://github.com/ghowland/papers/tree/main/papers/MATH/HOWL-MATH-8-2026